

The empirical recurrence for OEIS A253540, recovered from its tail

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Abstract

OEIS A253540 records “Empirical recurrence of order 86” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 550$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 94$. Its last coefficient is $c_{86} = 49000660836615138738725943470174622449664 \neq 0$, so no further tail satisfies anything shorter; any order-86 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253540 is “Number of $(n+2) \times (4+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum minus antidiagonal minimum nonincreasing horizontally and nondecreasing vertically.”. It has offset 1 and begins

119040, 2801200, 57759744, 1181976144, 23704397056, 477165435664, 9486518960384, 188451522370576, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:07:10, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 550$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 550$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 86: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 86.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1100$ exact terms from $n = 8$ on, it returns order 86 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	31	c_{23}	−341768443938513821696	c_{45}	5346563738920684248401786
c_2	−184	c_{24}	−827771057831588990976	c_{46}	−850515878201221538181526
c_3	−2148	c_{25}	−1391185289415162318848	c_{47}	1615745039877400194857865
c_4	41022	c_{26}	7377335948919726964736	c_{48}	8207646663669365832275242
c_5	−251642	c_{27}	−16123943098268839362560	c_{49}	−3471507311498026814695928
c_6	1049952	c_{28}	−460442670146849581629440	c_{50}	−2602676097013059137384666
c_7	−34478144	c_{29}	−706996747462373832130560	c_{51}	−43001490244636881124618309
c_8	87822795	c_{30}	−312139653553802568531968	c_{52}	122575898971188239913366602
c_9	1516825171	c_{31}	−15608244891239994134364160	c_{53}	577881161347777951834146648
c_{10}	−1193485072	c_{32}	3874372509345797963251712	c_{54}	489599673682372146409557115
c_{11}	83671585404	c_{33}	225641969494256906401742848	c_{55}	−162164544541332563073170973
c_{12}	−578779425824	c_{34}	1171437568819077997561643008	c_{56}	−534953675050378279263822825
c_{13}	4198050559216	c_{35}	8796384072426763727285518336	c_{57}	−530044903267646892442774063
c_{14}	−13495413067520	c_{36}	2746813553106781255723324032	c_{58}	1114794546611089796740407091
c_{15}	−146777693607360	c_{37}	5867434969807604484476502016	c_{59}	4151498897919290922185249586
c_{16}	−1671103999041152	c_{38}	−200055979205264852958280941568	c_{60}	2916496926985594534281548671
c_{17}	−13230193287482496	c_{39}	−548936220275908128342850666496	c_{61}	−713565113374203653729438725
c_{18}	39036369519984128	c_{40}	−679111465448173649021890461696	c_{62}	−1653990727826513643686939210
c_{19}	702935515294406656	c_{41}	−11310610079735553590254968504320	c_{63}	−816494182866693958363448181
c_{20}	3463457296755374080	c_{42}	−48429794590310033640013890060288	c_{64}	15340819947262831850417946673
c_{21}	5037914240715603968	c_{43}	54603713210741068431913564241920	c_{65}	2494651898935656093485142863
c_{22}	−59895799476330471424	c_{44}	562231241446100550193618580668416	c_{66}	2771764902110700916543654862

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 94$, and it is the recurrence OEIS A253540 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{86} - c_1x^{85} - \dots - c_{86}$. Its constant term is $-c_{86} = -49000660836615138738725943470174622449664$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 86 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 86, so $\deg q = 86$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 86, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 86, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 49000660836615138738725943470174622449664.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-86 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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